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EXXONMOBIL DATAWORKS CHALLENGE 2026

A Machine-Learning Proxy for Reservoir Simulation

Track T2: enabling fast, reliable history matching

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T2 Owner and Team Lead | 13 October 2026

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CONTEXT AND OBJECTIVE

Why a proxy model is needed

The challenge

History matching needs thousands of simulator runs to find parameters that reproduce real production. Each run is slow.

The T2 objective

Build a proxy that maps 4 reservoir parameters to the full production response, accurately enough to replace the simulator in the search.

The value

Thousands of candidate settings can be evaluated in seconds, enabling optimisation and uncertainty analysis by T3.

T1 Data and baseline
Done

T2 Proxy model
Done

T3 Optimise and forecast
Next

T4 Story and dashboard
Next

APPROACH

Problem framing and data design

Inputs (X)
4 parameters
fault transmissibility, porosity, permeability, aquifer volume

↓

Proxy
Machine-learning model trained to imitate the simulator

↓

Outputs (y)
123 values
oil, gas, water cumulative at 41 quarterly dates, 1998–2008

Case split (100 simulated cases)

Train Cases 1–70 · Fit and compare models (5-fold CV)

Validation Cases 71–85 · Confirm the chosen model

Test Cases 86–100 · Used once for final accuracy

Real field history is never a training target. It is used only to score matches.

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METHOD

Modelling pipeline

1 → 2 → 3 → 4 → 5

Scale

Inputs min-max to 0–1; each output stream divided by its own maximum

Compress

PCA per stream turns a 41-point curve into 6–7 shape coefficients

Learn

Gaussian Process (Matern 2.5) maps parameters to coefficients

Rebuild

Inverse PCA and rescale back to STB and MSCF

Enforce physics

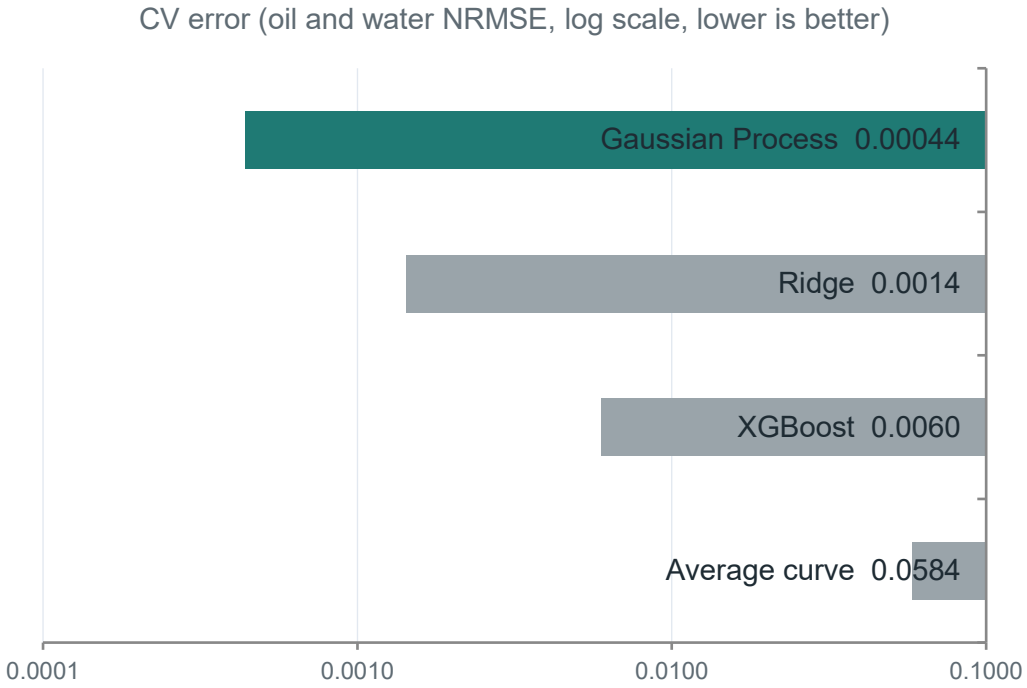
No negatives; cumulative never decreases

Leakage control: steps 1–3 are refitted inside every one of the 5 cross-validation folds, using only that fold's training cases. Test cases were evaluated once, after all choices were frozen.

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RESULTS – 5-FOLD CROSS-VALIDATION

Gaussian Process selected as the proxy



~130×

lower error than the average-curve reference

~13×

lower error than XGBoost on 70 smooth curves

3×

lower error than quadratic Ridge

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RESULTS – VALIDATION AND ONE-TIME TEST

Final accuracy on unseen cases

0.05%
mean error at Jan 2008

1.000
ranking correlation

3.2 s
for 50,000 predictions

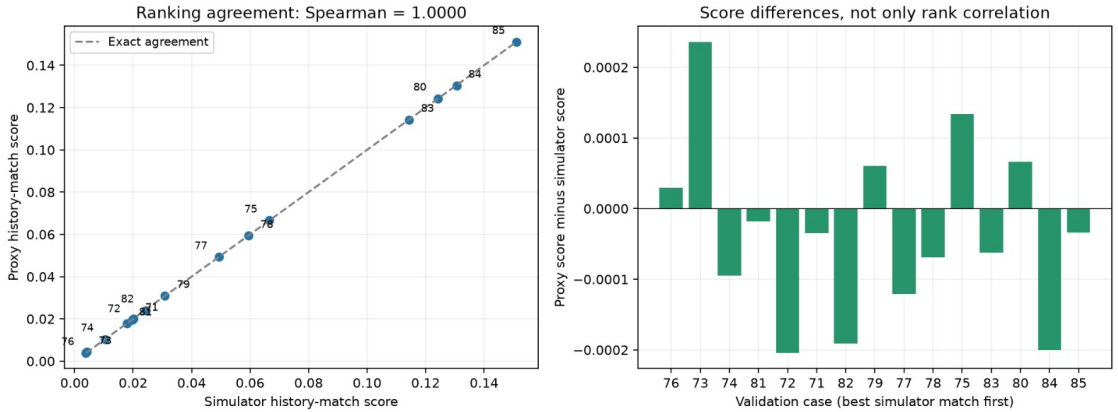
36
automated tests passing

NRMSE (target < 0.002)	Oil	Gas	Water
Cross-validation (cases 1–70)	0.00028	0.00026	0.00060
Validation (cases 71–85)	0.00018	0.00016	0.00037
Final test (cases 86–100)	**0.00022**	**0.00020**	**0.00046**
Mean error at Jan 2008 (test)	0.04%	0.04%	0.07%

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RESULTS – FITNESS FOR T3

Proxy ranks matches like the simulator



Both scores compare curves with RAMP history using unchanged T1 weights. This checks score/rank fidelity, not improvement over Case 6.

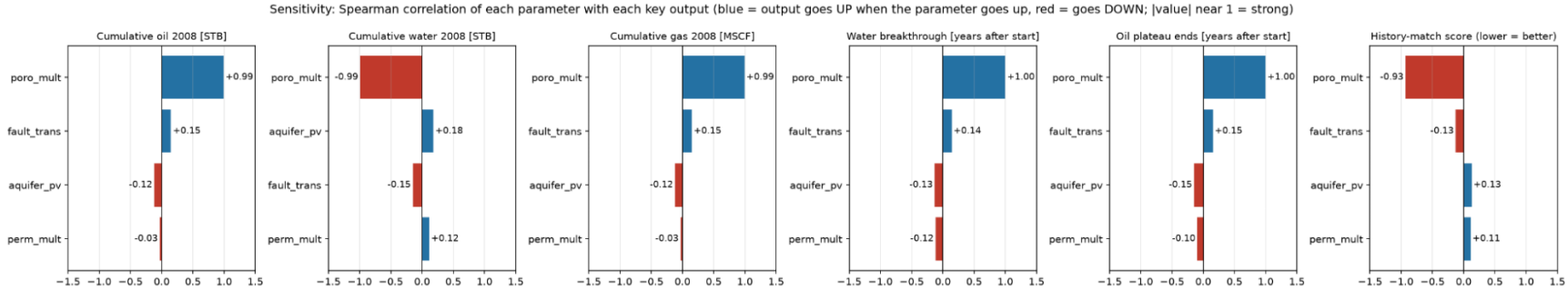
Validation cases 71–85: simulator vs proxy history-match scores

- Spearman = 1.000 on validation and test
- Top 10 overlap 10/10; same best case in each split
- Mean score error 0.0002, which is small relative to the spread of scores
- Near-ties (gaps < 0.0003) are treated as equivalent

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INSIGHTS

Supporting insights from the data



Spearman correlation of each parameter with key outputs (T1 sensitivity analysis, cases 1–85)

Porosity dominates

Correlation about 0.99 with oil, water and breakthrough timing in 1998–2008.

Non-uniqueness

Fault, permeability and aquifer are weakly constrained by history, so several settings match equally well.

Gas tracks oil

Gas is about 0.363 × oil; the proxy preserves this within 0.08% without being told.

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RISK AND TRANSPARENCY

Limitations and how they are controlled

Limitation	Evidence	Control
Uncertainty bands too narrow	83–89% coverage vs 95% nominal	Reported as uncalibrated; not used as risk bounds
Near-tie resolution	Score error up to ~0.0003 near good matches	Treat close scores as ties; choose a diverse top 10
Valid only inside parameter range	GP reverts to average outside data	Search restricted to declared bounds; warnings
History period only (1998–2008)	No data after Jan 2008	2028 forecast handled by T3's separate method
Late-time rate sensitivity	Final-quarter rate error up to 2.9%	T3 tests sensitivity of decline-curve start

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HANDOVER AND NEXT STEPS

Ready for T3 optimisation

```
from src.proxy import predict, param_bounds

bounds = param_bounds()
y = predict(X)   # (n, 4) -> (n, 123)

Saved model: models/proxy.joblib
```

1. Push t2-proxy branch and open pull request for T3 review
2. T3: search within bounds, select a diverse top 10
3. T3: forecast 2008–2028 and derive P10/P50/P90
4. Code submission: 16 October, 3 PM
5. Slides, Power BI and video: 20 October, 3 PM

Proxy: 0.05% error at 2008 · perfect ranking on held-out cases · 50,000 predictions in 3.2 s